

Pure Substances and Mixtures — Paper D

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must say whether the atoms are **chemically joined** or **not joined**, and whether there is **one kind of particle** or more than one.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe substances or experiments you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- "Pure" on this paper always has its chemical meaning: one substance only.

SECTION 1 – IDENTIFYING FROM DATA

1. A data book lists the melting points of four pure white solids:

urea 133 °C • benzoic acid 122 °C • stearic acid 69 °C •
naphthalene 80 °C

Three unlabelled samples are tested. Sample 1 melts at 80 °C. Sample 2 melts between 61 and 67 °C. Sample 3 melts between 118 and 122 °C.

(a) Identify sample 1 and explain how you know it is pure. [2]

(b) Sample 2 is most likely an impure form of which solid? Give two reasons. [2]

(c) A student says sample 3 must be pure benzoic acid because "it reaches 122 °C". Explain why this is wrong. [2]

2. Hydrogen is a gas that burns. Oxygen is a gas that makes things burn faster. [6]
Water, which is made from only these two elements, puts fires out. A student says this proves water is a mixture of hydrogen and oxygen.

Explain why water is a **compound** and not a mixture, and why it is nevertheless a **pure substance**. Your answer should refer to how the atoms are held, the ratio of the elements, the properties of water compared with its elements, and what it would take to separate the hydrogen from the oxygen.

SECTION 3 – APPLYING THE IDEAS

3. A jar contains a mixture of hydrogen gas and oxygen gas. A spark is put in and there is a bang; afterwards the jar contains only water droplets.
- (a) State whether the contents of the jar before the spark were pure or a mixture, and why. [2]

- (b) Explain what the spark caused to happen to the particles. [2]

4. Give **three** differences between the mixture of iron and sulfur before heating and the compound iron sulfide formed after heating. Each difference must say something about both. [3]

5. A student draws a "mixture of two elements" as pairs of one black circle joined to one white circle, scattered across the box.

(a) State what the student has actually drawn. [1]

(b) Describe how the diagram should be corrected, and explain why the correction matters. [2]

6. Two students wrote the following in a test. Neither answer is correct. [5]

(i) "A pure substance is one that occurs naturally and has had nothing added to it."

(ii) "An impure sample melts at a lower temperature, so the melting point simply moves down."

For each one, explain the mistake and write a correct version.

- 7.** Explain why the melting point of a pure substance can be used to identify it, but the melting range of a mixture cannot. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.